

Research paper

Advancing building energy efficiency: A deep learning approach to early-stage prediction of residential electric consumption

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ABSTRACT

Enhancing building energy efficiency underscores the critical need for innovative predictive models to mitigate environmental issues from high energy consumption, especially in residential areas with air-conditioning and heating ventilation systems. This study introduces the use of Long Short-Term Memory (LSTM) networks for early prediction of residential electric consumption, representing a significant innovation in the field. Unlike traditional Deep Neural Network (DNN) and Artificial Neural Network (ANN) models, Long Short-Term Memory networks efficiently process time-series data, predicting future energy usage with unmatched accuracy. The Long Short-Term Memory model exhibited superior training efficiency, requiring only 2.69 s for over 500 test cases, outperforming Deep Neural Network and Artificial Neural Network models, which took 5.26 and 3.88 s, respectively. Its predictive accuracy, evidenced by an R-squared value of 0.97, surpasses the 0.95 and 0.92 of Deep Neural Network and Artificial Neural Network models, respectively. This breakthrough enables accurate predictions of annual energy usage before construction starts and aids in identifying energy efficiency improvements early in the design process. Applying Long Short-Term Memory networks in this context marks a substantial advancement in predictive modeling for building energy consumption, equipping architects and engineers with a vital tool for designing energy-efficient buildings from the beginning. The innovation and quantitatively proven effectiveness of the Long Short-Term Memory model highlight its potential to revolutionize early-stage building design strategies, filling a crucial gap in the existing literature.

1. Introduction

1.1. Motivation and Background

Energy used in buildings is now a considerable portion of overall energy usage due to global population growth and economic development (Hu et al., 2021). Reports state that buildings account for 38 % of carbon dioxide (CO₂) emissions and 40 % of global energy consumption. It is predicted that, by 2030, global energy consumption will increase by more than 50 % (Mawson and Hughes, 2020). Buildings consume a significant amount of energy, leading to various environmental issues that adversely affect human existence. The enormous energy consumption and its harmful impacts emphasize the importance of understanding electric energy efficiency and identifying potential

improvements in newly designed buildings. This concept is widely accepted, and numerous governments have introduced various laws, guidelines, and incentives to minimize electric energy use in residential and commercial premises (Amasyali and El-Gohary, 2021). Using machine learning (ML), new advancements have been made in research on energy efficiency in buildings (Fan et al., 2021). Forecasting the electric energy utilization of buildings is crucial for energy efficiency and improved decision-making to reduce power demand. However, predicting power demand remains challenging due to various factors, including weather, the physical characteristics of a building, and the power requirements of residents (Runge and Zmeureanu, 2019). The construction of electric energy forecasting models is typically divided into two methodologies: forward and data-driven strategies. Forward models, often referred to as physics-based modeling approaches,

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commonly require a comprehensive array of explanatory inputs related to the building's architecture and its environmental context. The surroundings include information on the HVAC system, insulation thickness, thermal properties, internal occupancy loads, and solar data (Tardioli et al., 2015). These models require several frequently unattainable parameters. Due to the time required and the lack of necessary information, these technologies are considered inefficient.

The mathematical models and their accurate data are the foundations of data-driven models to make accurate decisions. Earlier, many autoregressive and stochastic time-series models were used to estimate the daily and weekly electricity loads. Recently, with statistical ML approaches, prediction tasks focusing on economics have implemented ML algorithms (Olu-Ajayi et al., 2022a). Predicting the energy usage of buildings before construction begins is crucial for reducing the number of inefficient buildings that consume significant amounts of electricity. The best approach for achieving the desired prediction result is the ML methodology due to its high performance in less time (Kim and Cho, 2019). Accurate prediction of a building's energy usage is intended to decrease energy usage by 10–30 % (McNeil et al., 2019). Therefore, a persistent mechanism to predict electricity consumption is important for well-organized buildings. The development of data-driven approaches has led to successful energy estimation outcomes.

1.2. Related literature

In addition, predicting building energy consumption results in lower greenhouse gas emissions, more energy-efficient buildings, and increased financial savings (Bagnasco et al., 2015). ML algorithms have been widely implemented in building electric energy forecast models. These algorithms include support vector machines (SVM), decision trees, and artificial neural networks (ANN) (Bourhane et al., 2020). These ML algorithms have proven accurate for a large dataset of a single building, but the model provides inaccurate predictions when the dataset contains more buildings (Wang et al., 2020a). An ANN emulates the function of the human brain to solve big data nonlinear computational problems (Liu et al., 2020). ANN is a practical approach for solving nonlinear problems, especially with big data, and enables the neural network to effectively train the model with sufficient data.

ANN-based energy prediction models have produced good results when applied to a dataset of a 6-story building with two blocks (Li et al., 2009). SVM was implemented to estimate the hourly cooling load of a building based on meteorological data and was proven to perform better than the neural network models with less error (Dong et al., 2021). A comparative study of ANN and SVM models was conducted on a dataset collected from 507 buildings. The dataset contained structural and meteorological data, such as building type, floor area, humidity, and atmospheric pressure. The results showed that ANN is superior to SVM with a lower root-mean-squared error. The decision tree algorithm is utilized for forecasting interpretable data and for better classification. The neural network and decision tree performed better than the regression method (Yu et al., 2010). In addition, technological transfer toward the ensemble approach produces enhanced results. The ensemble model comprises numerous ML algorithms stacked in a desirable order to perform better than a single algorithm (Kim et al., 2024). Random forest (RF), decision trees (DT), and gradient boosting are some of the algorithms used in the ensemble approach due to their data stability (Zhong et al., 2019). A comparison between the ensemble approach and ANN revealed that the ensemble approach yielded superior outcomes in comparison to ANN (Olu-Ajayi et al., 2022b). The efficacy of machine learning (ML) algorithms is contingent upon the accessibility of datasets and input features. A particular investigation implemented various ML algorithms, including SVM (Ma et al., 2023), DT, RF (Cheng et al., 2022), boosting algorithms, K-Nearest Neighbor (KNN) (Hong et al., 2022), Gaussian Process (GP), and Multi-Layer Perceptron (MLP) (Afzal et al., 2023), to forecast energy utilization during the design phase of buildings. Among these, the Gradient

Boosting algorithm was found to achieve a prediction accuracy of 67 % (Kim et al., 2021). Although various structural and meteorological factors are considered for predicting energy consumption, it is necessary to identify the significant factors.

Several methods have been used to predict residential electric consumption, each with its strengths and weaknesses. ARIMA is a traditional time-series forecasting method widely used for predicting energy consumption (Yang et al., 2022). While effective for short-term predictions, ARIMA struggles with long-term predictions and handling missing data. It assumes linear relationships and may not capture complex interactions between climatic and building data. Support Vector Regression (SVR) is effective at handling non-linear relationships in data. SVR works well with smaller datasets, but its accuracy decreases with larger and more complex time-series data (Chen et al., 2023). Random Forest Regression (RFR) can handle high-dimensional data and capture complex interactions between features (Liu et al., 2023). This capability is crucial given the variability and potential gaps in climatic and energy usage data.

Gradient Boosting Machines (GBM) are known for high accuracy and the ability to handle various data types (Wang and Zhang, 2023). Despite the strengths of RFR and GBM, these algorithms struggle with sequential data due to the non-recurrent structure. Traditional methods like ARIMA, SVR, RFR, and GBM face challenges in handling sequential data, capturing long-term dependencies, managing missing data, and modeling complex interactions. These limitations make these methods less suitable for predicting residential electric consumption based on the given time-series data.

Tuning the hyperparameters of the ML algorithm also boosts the performance of the predictive model. Training massive amounts of data for statistical models beyond their capability is expensive, and scalability causes performance degradation. Neural networks perform predictions well with nonlinear datasets and effectively handle big data. DNN and deep learning algorithms perform well on small datasets (He, 2017). LSTM, based on the sequence-to-sequence model, was utilized to predict minute-by-minute load data. Convolutional neural network-based architectures are used for imputing missing data because real-time datasets are mostly incomplete. A generative adversarial imputation network (GAN) can handle high-dimensional complex data with a probability distribution. GAN is implemented to estimate the load produced by the wind on tall buildings (Jankovic et al., 2023). KNN and GAN can achieve greater accuracy with less error. A deep convolutional GAN (DC-GAN) model was used to forecast the pressure constants at various periods on tall structures. The model promised high prediction accuracy by imputing lost data in the dataset (Faiq et al., 2023).

The power usage of campus buildings is assessed by employing LSTM networks, with a particular focus on incorporating forthcoming weather data into the predictive model. The efficiency of the methods depends on the historical data considered for training the model. LSTM, GRU, and Drop GRU models are employed for predicting power utilization (Mahjoub et al., 2022). Through a comprehensive comparative analysis, the predictive accuracy and computational efficiency of these deep learning models are examined, providing valuable insights into their applicability for energy consumption forecasting tasks. The periodicity of consumption patterns is leveraged for predicting energy usage (Wang et al., 2020b). The study investigates how LSTM excels in capturing recurring trends and delivering precise forecasts over extended timeframes, providing valuable insights into its potential for forecasting energy consumption across diverse scenarios. An integrated method is adopted, merging Convolutional Neural Network (CNN) and Bi-Directional LSTM networks, aiming to produce accurate energy predictions (Ullah et al., 2020). The CNN primarily serves to extract features from input data, while the Multi-Layer Bi-Directional LSTM networks are employed to determine the temporal dependencies and bidirectional information embedded in the sequence data. Integration of LSTM networks with various neural network architectures is applied to improve the accuracy of power usage predictions (Somu et al., 2021).

This model integrates pertinent features and historical data to anticipate household energy consumption patterns, employing methodologies such as data preprocessing, model training, and evaluation to fine-tune forecasting precision.

The LSTM model has demonstrated superior predictive accuracy and better handling of missing data (Ju and Liu, 2021). The temporal nature of energy consumption data benefits greatly from LSTM’s ability to capture time-based patterns and trends (Li and Jiang, 2023). While traditional machine learning algorithms and neural network architectures have their merits, the LSTM model’s ability to handle sequential data, capture long-term dependencies, manage missing data, and model complex interactions makes it the optimal choice for predicting residential electric consumption based on the given time-series data.

1.3. Novel contributions

The development of an accurate prediction model for energy consumption is essential during the pre-construction phase of buildings. The predictive model helps to save energy, optimize building design, and support environmental energy conservation. This study presents a technique to predict annual electricity consumption by applying the Long Short-Term Memory (LSTM) algorithm. The developed models estimate the electricity consumption of a building during the construction phase for future periods based on the heating and cooling loads of the building and outdoor climate-based data. The highlights of the proposed work are:

- Applied adversarial training to enhance robustness in predicting building energy consumption.
- Developed an LSTM model for accurate annual electricity consumption forecasts using climatic data.
- Integrated dynamic input handling and feature engineering in LSTM for improved performance.
- Validated LSTM’s superiority over Deep Neural Network (DNN) and Artificial Neural Network (ANN) models through comprehensive analysis.

2. Materials and methods

2.1. Dataset description

Climatic (meteorological) data and building metadata were used to develop the model. The Ministry of Housing, Communities and Local Government (MHCLG) repository was mined for the building-related dataset. This included power consumption information and metadata of 5000 residential buildings in the U.K (Yuvaraj et al., 2020). For a more complete and precise comparison, we selected data from 600 domestic (residential) buildings from each of 10 geographical areas

(Darlington, Yorkshire, Halton, Blackburn, Blackpool, Hartlepool, Redcar Cleveland, Hull, Middlesbrough, and Warrington). The types of buildings included maisonettes, bungalows, flats, and houses. Only the variables that can be discovered and altered during the design phase were included in the building metadata. It provided the number of habitable rooms and a description of the roofs and walls and was classified as independent parameters, whereas the energy data were regarded as dependent parameters. The 2020 energy usage of each building was included in the dataset. Input parameters, such as the description of the walls and windows, are significant factors because they affect the energy usage of a building (MHCLG open data, 2022). Climatic data included weather-related attributes, such as climate, wind speed, and pressure. This data was collected from the Meteosat repository (The Weather’s record keeper, 2022) from January 1 to December 31, 2020, based on the average daily weather for ten residential building areas. Climatic data were classified as exterior data, and the building data were classified as interior. Table 1 shows the selected parameters for the complete repository. The table consists of six attributes with thirteen different parameters.

The selection of the 13 parameters for our model was guided by their proven impact on building energy efficiency, relevance to early-stage design considerations, and empirical validation through preliminary testing. These parameters encompass critical environmental influences and building characteristics, ensuring a comprehensive approach to predicting residential electric consumption. The choice to limit the model to these parameters was based on achieving an optimal balance between predictive accuracy and practical applicability, supported by literature review and data availability. Based on the complete dataset, various statistical measures were calculated, including the standard deviation, mean, highest, and lowest values of each parameter. The input parameters included temperature, wind speed, pressure, wall, roof, surface area, relative compactness, glazing area distribution, overall elevation, orientation, and heating and cooling loads. Using these input parameters, the electric energy consumption was predicted.

2.2. Data preparation

Data preprocessing is time-consuming and computationally expensive, but it is crucial for identifying inconsistent or incorrect data that can lead to errors in the analysis. Data merging of numerous datasets was performed to form a combined extensive dataset. The standard variable (postal address) was used to combine the building dataset with meteorological information, matching each building dataset with its corresponding meteorological dataset. Missing data and outliers were treated throughout the data-cleaning process. The unprocessed data collected from the buildings included categorical information on aspects such as wall and window energy efficiency. Data normalization is a widely used technique for preprocessing to remove the impact of

Table 1
Dataset description.

Parameters	Type of data	Mean	Standard Deviation	Minimum	Maximum
Temperature (°C)	Continuous	13.660	6.182	-0.060	32.400
Wind speed (km/h)	Continuous	3.581	1.694	0.200	9.960
Pressure (Pa)	Continuous	1014.127	11.073	979.250	1040.920
Relative compactness	Continuous	0.763	0.105	0.620	0.980
Surface area (m ²)	Continuous	671.708	88.086	514.500	808.500
Wall area (m ²)	Continuous	318.500	43.626	245.000	416.500
Roof area (m ²)	Continuous	176.604	45.165	110.250	220.500
Overall height (m)	Continuous	5.250	1.751	3.500	7.000
Orientation (north, east, west, south)	Discrete	3.500	1.118	2.000	5.000
Glazing area (of the floor)	Discrete	0.234	0.133	0.000	0.400
Glazing area distribution (variance)	Discrete	2.812	1.550	0.000	5.000
Heating load (kWh/m ²)	Continuous	22.307	10.090	6.010	43.100
Cooling load (kWh/m ²)	Continuous	24.587	9.513	10.900	48.030
Energy consumption (kWh/m ²)	Continuous	Output			

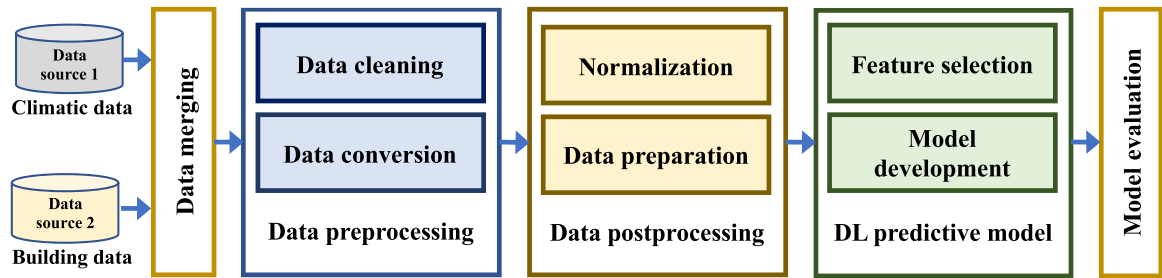


Fig. 1. General working of a deep learning predictive model.

dimensions because many attributes frequently have unrelated sizes. A normalization process was used to scale individual data into a unit norm (Kim et al., 2020; Malekian, and Chitsaz, 2021).

Fig. 1 depicts a comprehensive overview of the deep learning framework. The first step is data merging, wherein various types of input data were consolidated to create a unified data source (dataset). Subsequently, data preprocessing involved two critical stages: data cleaning and data conversion. Following data preprocessing, the dataset underwent normalization and preparation for model training during post-processing. In the context of a deep learning model, principal component analysis was employed to select the primary features of the dataset before initiating the prediction training phase. Finally, model evaluation ensued, involving a comparative assessment between the predicted and actual results to assess the efficiency of the model.

2.3. Proposed model

The actual purpose of the proposed technique is to involve deep learning algorithms in developing an intelligent predictive model. While working with sensors and continuous time-series data, failures and missing data can occur in the dataset. Some columns may not have any values. A model operates well and obtains high prediction accuracy when it has sufficient enriched data. A sufficient amount of high-quality data input can decrease training errors and boost the overall efficiency of the model. In this proposed work, the dataset comprised three different types of data: historical data from newly constructed buildings, climatic data, and building data. These data were pre-processed using various strategies discussed in the data preparation section. To validate the generalizability and applicability of the LSTM model, additional datasets from various building types and geographic locations were

utilized. These datasets included residential buildings with different construction styles, sizes, and climatic conditions. The prepared data were then used to train the prediction models: LSTM, DNN, and ANN. After predicting annual electricity consumption using the three models, the results were compared for model validation.

Fig. 2 illustrates the process flow of the proposed predictive model. Initially, the first step involves the collection of data pertinent to energy consumption prediction from various sources. Following this initial collection, the data is categorized in the second step into either continuous or discrete types. The third step details the preprocessing activities conducted on the collected data. It is noteworthy that during the data acquisition phase, particularly from sensor inputs and continuous time series, instances of data anomalies and missing entries may arise. In the fourth step, an evaluation of the data preprocessing and enrichment techniques is conducted to ensure data quality prior to model development. Subsequently, the fifth step introduces the application of an LSTM-based approach for predicting annual electricity consumption. Additionally, the efficacy of the proposed LSTM model will be assessed by conducting a comparative analysis with two other deep learning models.

3. Smart prediction model

This research uniquely applies adversarial training techniques to the context of energy consumption prediction for residential buildings. Adversarial training enhances the robustness and reliability of the predictive model by effectively addressing common data quality issues such as missing and faulty sensor data, which are prevalent in real-world datasets. This approach has not been extensively explored in building energy prediction, marking a significant innovation in the methodology.

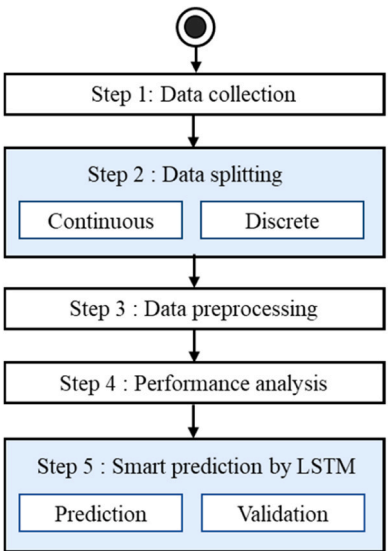


Fig. 2. Workflow of a proposed predictive model.

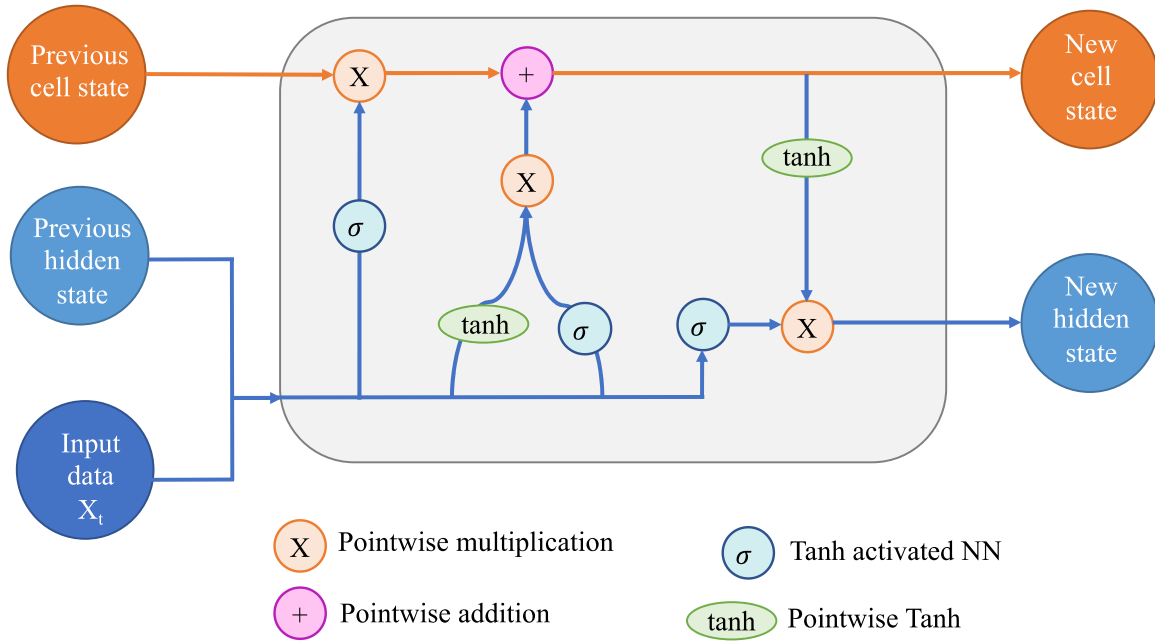


Fig. 3. Overview of LSTM model.

Additionally, an advanced data prediction framework was developed leveraging Long Short-Term Memory (LSTM) networks to forecast annual electricity consumption based on climatic parameters. The LSTM model's ability to capture long-term dependencies in time-series data provides significant advantages over traditional deep neural networks (DNN) and artificial neural networks (ANN). Extensive validation results demonstrate that the LSTM model achieves superior predictive accuracy, particularly in scenarios with incomplete or imperfect data.

3.1. LSTM

Long Short-Term Memory (LSTM) networks are highly effective for learning and preserving extended dependencies within sequential data. LSTM networks outperform feedforward networks due to the capacity to process new data points using previously stored sequence information. LSTM networks generate output through computations that involve input data, the current hidden state, and the internal memory cell. The cell state enables the capture of extended dependencies, updated through gating mechanisms like forget, input, and output gates. These gates regulate information flow, maintaining or discarding information based on its relevance. The output, influenced by the cell state, provides contextual and historical insights from previous time steps. Typically, the output is computed based on the current hidden state, which holds essential information for decision-making or prediction within the sequence (Karthic and Manoj Kumar, 2022) Fig. 3.

The input gate assumes a pivotal role in orchestrating the incorporation of novel data originating from both the present input and the preceding hidden state into the memory cell. It functions as a regulatory mechanism governing the transfer of data into the LSTM cell, thereby wielding significant control over the network's ability to assimilate insights gleaned from incoming data sequences, while safeguarding critical information over extended time spans. The computation of input gate is represented as:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (1)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (2)$$

where i_t indicate the state of input gate; h_{t-1} represents the previous hidden state; $\tilde{C}_t$ is the present cell state. The forget gate determines which cell state bits are useful given the initial hidden state and new input data. The forget gate vector consists of values ranging between 0 and 1. A value approaching 1 signifies the importance of retaining the corresponding element in the memory cell state, whereas a value approaching 0 suggests that the information should be discarded. The forget gate first obtains the input $x(t)$ and output $h(t-1)$, then multiplies its weight matrix point-wise with sigmoid activation to provide the probability scores. The forget gate f_t is represented as:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (3)$$

The output gate in LSTM networks selectively controls the flow of information from the current memory cell state to the output, thereby exerting a significant influence on the final prediction or output generated by the LSTM network. The output gate functions by taking the current hidden state h_t and the current input x_t as inputs. These inputs are concatenated and then subjected to a sigmoid activation function to produce the output gate vector O_t . Our output was complete after the last tanh multiplication to ensure that the values range from [-1,1].

$$O_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (4)$$

$$h_t = O_t * \tanh(C_t) \quad (5)$$

where O_t represents the output gate and the hidden state is represented as h_t .

Algorithm 1. Working of LSTM

Input: Normalized input data

Output: Forecasted yearly electricity usage of the residential building

1. Randomly initialize weights for the LSTM gates: W_i, W_f, W_c, W_o
2. Set the recurrent data weights for LSTM gates: R_i, R_f, R_c, R_o
3. Choose the peephole parameter: $\mathcal{V} \in \mathbb{R}^N$
4. Choose the bias parameter concerning $\mathbb{R}^N$: b_i, b_f, b_c, b_o
5. Determine the input gate value $i_t = \sigma(W_{ix_t} + R_i h_{t-1} + b_i)$ concerning t .
6. Determine the forget gate value $f_t = \sigma(W_{fx_t} + R_f h_{t-1} + b_f)$ concerning t .
7. Determine the Input for the present cell nodes.

$$\tilde{C}_t = \tanh(W_{cx_t} + R_c h_{t-1} + c)$$

$$C = i_t \odot \tilde{C}_t + f_t \odot C_{t-1}$$

8. Determine the value of output gate $o_t = \sigma(W_{ox_t} + R_o h_{t-1} + b_o)$
9. $h = o_t \odot \tanh(C_t)$ is the result of the node.

Algorithm 1 shows the sequence of steps involved in determining the annual electricity consumption with the support of LSTM gates.

The input variables for predicting annual electricity consumption included historical data from newly constructed buildings, climatic data (temperature, humidity, solar radiation, wind speed), and building characteristics (floor area, number of floors, insulation type, window-to-wall ratio). The LSTM model architecture comprised an input layer, two LSTM layers with 50 units each (with 0.2 dropout), a dense layer with 25 neurons, and a single neuron output layer. Hyperparameter tuning optimized LSTM units, dropout rate, batch size, and learning rate using grid search with cross-validation. The dataset was split 80:20 for training and testing, with further division for validation. The model was trained using the Adam optimizer and mean squared error (MSE) as the loss function. Early stopping was implemented to prevent overfitting, with a patience of 10 epochs. The model's performance was evaluated using metrics such as mean squared deviation (MSD), mean absolute error (MAE), root-mean-square error (RMSE), and R-squared (R^2) value.

The proposed LSTM architecture incorporates several innovative elements that significantly enhance its predictive capabilities. One of the key innovations is the dynamic input handling feature. This allows the LSTM model to manage varying input lengths effectively, which is particularly beneficial for time-series data that may have missing entries or inconsistent time intervals. This adaptability improves the model's

robustness and accuracy, ensuring reliable predictions even when data quality is compromised.

Another critical advancement is the enhanced feature engineering process. This process involves selecting and prioritizing the most relevant climatic parameters, thereby reducing noise and improving the quality of the input data fed into the LSTM model. By focusing on the most impactful variables, the model's performance is significantly boosted, resulting in more accurate and reliable predictions. Furthermore, the integration of adversarial training with the LSTM architecture marks a significant innovation. Adversarial training enhances the model's resilience to adversarial examples and noise, which are common issues in real-world sensor data. This integration ensures that the model remains accurate and reliable even in the presence of data imperfections, thereby increasing its practical utility in real-world applications.

3.2. DNN

The prediction model uses various types of deep neural network (DNN) components. The input features of the load-forecasting model come in a wide variety of forms. A suitable neural network component uses a certain input, develops a deep representation, and extracts high-quality information from the input training dataset. The historical electric load sequence is the most essential input, providing valuable

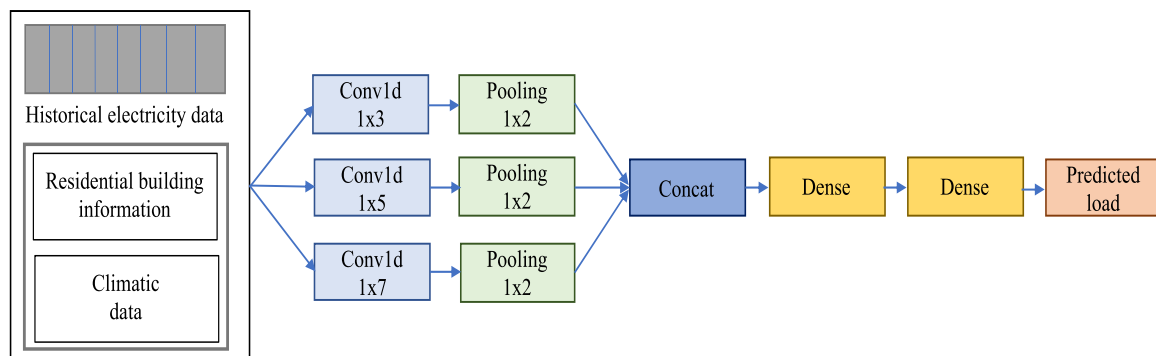


Fig. 4. Working of the deep neural network.

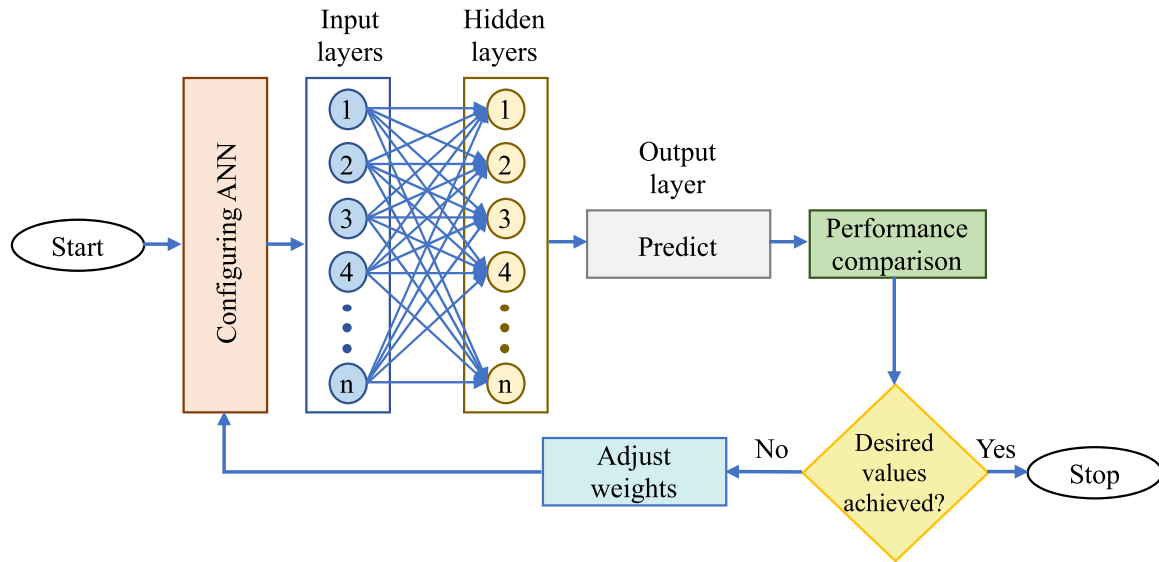


Fig. 5. Working of artificial neural network (ANN) model.

data for forecasting future load demand. The model processed the historical load data using parallel CNN components.

The features extracted from the raw data typically manifest as statistical metrics, such as the maximum electrical demand on the previous day and the demand during the same time on the last day. In the first layer of a DNN model, CNNs are typically employed for feature learning and extraction. Additionally, they function as filters and can transform the input data using kernels with nearly connected receptive fields. Therefore, they can learn a variety of features from unprocessed inputs. To obtain a range of features for future load prediction, a CNN was used in this model to convert the historical load series. The DNN model can be trained using the features extracted from the input sequence by several simultaneous CNN components with various filter sizes.

The historical load sequence, along with key features retrieved by the CNN mechanisms, represents time series data characterized by complexity and dynamism. Supplementary factors, including weather information, off days, and precise time and day information, were integrated into the forecasting model along with the historical load series. The DNN model received all concatenated characteristics as input, as illustrated in Fig. 4. This procedure enables the model to discern abstract features from raw inputs more effectively.

3.3. ANN

An ANN involves mathematical models and complex computations, similar to the human brain. The architecture of an ANN is similar to the biological structure of the human nervous system. Like the neurons of the human brain, an ANN comprises a computational unit of neurons that work in a non-linear form. A neuron contains an input, an output, and several intermediate hidden layers. A neural network framework consists of numerous neurons interconnected with one another. The

ANN model uses several input variables through the input layer, performs regression at the hidden layer utilizing an activation function, and produces the result through the output layer. Each input variable is assigned a weight provided by the ANN framework based on the learning and cost functions used in the hidden layer. Fig. 5 depicts the working model of the ANN.

An ANN is a multi-layer feed-forward network in which the input layer of a neuron receives data from the output layer of the previous neuron. The variables at the input layer are combined linearly using a weighted linear combination, as stated in Eq. 6, and are modified using a nonlinear sigmoidal activation function at the hidden layer, as in Eq. 7.

$$O_j = B_j + \sum_{i=1}^n w_{ij} \cdot X_i \quad (6)$$

$$O(x) = \frac{1}{1 + e^{-x}} \quad (7)$$

Table 2

Performance metrics.

Performance Metrics	Formula
R-squared (R^2)	$R^2 = 1 - \frac{\sum_m (A_m - P_m)^2}{\sum_m (A_m - \bar{A}_m)^2}$
Mean absolute error (MAE)	$MAE = \frac{1}{N} * \sum A_m - P_m $
Root mean square error (RMSE)	$RMSE = \sqrt{\frac{\sum_{i=1}^N \ A_m - P_m\ }{N}}$
Mean standard deviation (MSD)	$MSD = \frac{1}{N} \sum_{i=1}^N (A_m - P_m)^2$

The performance was then compared utilizing the RMSE and MSD. Table 3 shows the comparison of the three predictive models.

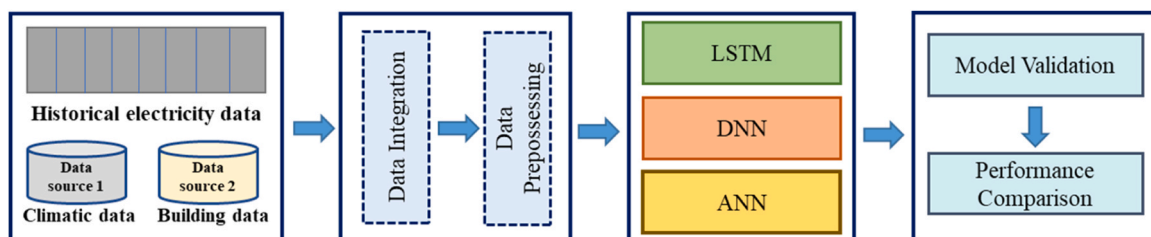


Fig. 6. Data prediction model.

Table 3
Performance comparison.

Performance Parameter	LSTM	DNN	ANN
Training time	2.69	5.26	3.88
R-squared	0.97	0.95	0.92
Mean absolute error	0.90	0.92	0.96
Root mean square error	1.01	1.16	1.2
Mean standard deviation	0.98	1.34	1.45

where X represents the set of input variables from $i = 1$ to n ; O denotes the output variable; B is the biasing variable; w is the assumed weight of input i ; j is the corresponding output variable for input i ; $O(x)$ is the sigmoidal nonlinear activation function.

In the feedback recurrent ANN, the output layers feedback on the result to the input layer or other hidden layers, forming a loop. The ANN automatically learns the relationship between the input features and the output by analyzing the input via sampling and minimizing the difference between the input and output. The input feature data pass through several nonlinear hidden layers at each neuron, and input data aggregation from the previous neuron is performed at the next neuron. The iteration is repeated by adjusting the weights until the model yields the desired outcome. The outcome of the ANN model is measured based on the number of hidden layers. An increase in the number of hidden layers enhances the performance and complexity of the model. The proposed

work used an LSTM model to estimate electricity consumption, and the outcome was compared with that of DNN and ANN models.

4. Results and discussion

This section elucidates the empirical findings from the deployment of the proposed deep-learning-based intelligent prediction model, alongside a comparative evaluation with other deep learning models. The outcomes demonstrated notable correlations among the evaluated models, with the LSTM model specifically designed to forecast annual electricity usage. The efficacy of the LSTM model was established through comparative analysis with other deep learning architectures, including DNN and ANN, based on parameters such as mean square deviation (MSD), mean absolute error (MAE), and root-mean-square error (RMSE). Values approaching zero, and an R-squared value nearing one, indicated superior performance. Among the models assessed, the LSTM model emerged as the most proficient in estimating annual energy consumption within this study. While the DNN model, renowned for its accuracy in analyzing extensive datasets, also exhibited commendable performance, it did not surpass the LSTM model in effectiveness.

Fig. 6 depicts the complete process of the data prediction models. The dataset comprises three different types of data: historical data obtained from newly constructed buildings, and climatic and building data. The data were then pre-processed using the different strategies discussed in the data preparation section. The prepared data were then transferred to the prediction models: LSTM, DNN, and ANN. After predicting annual electricity consumption using the three models, the results were compared for model validation.

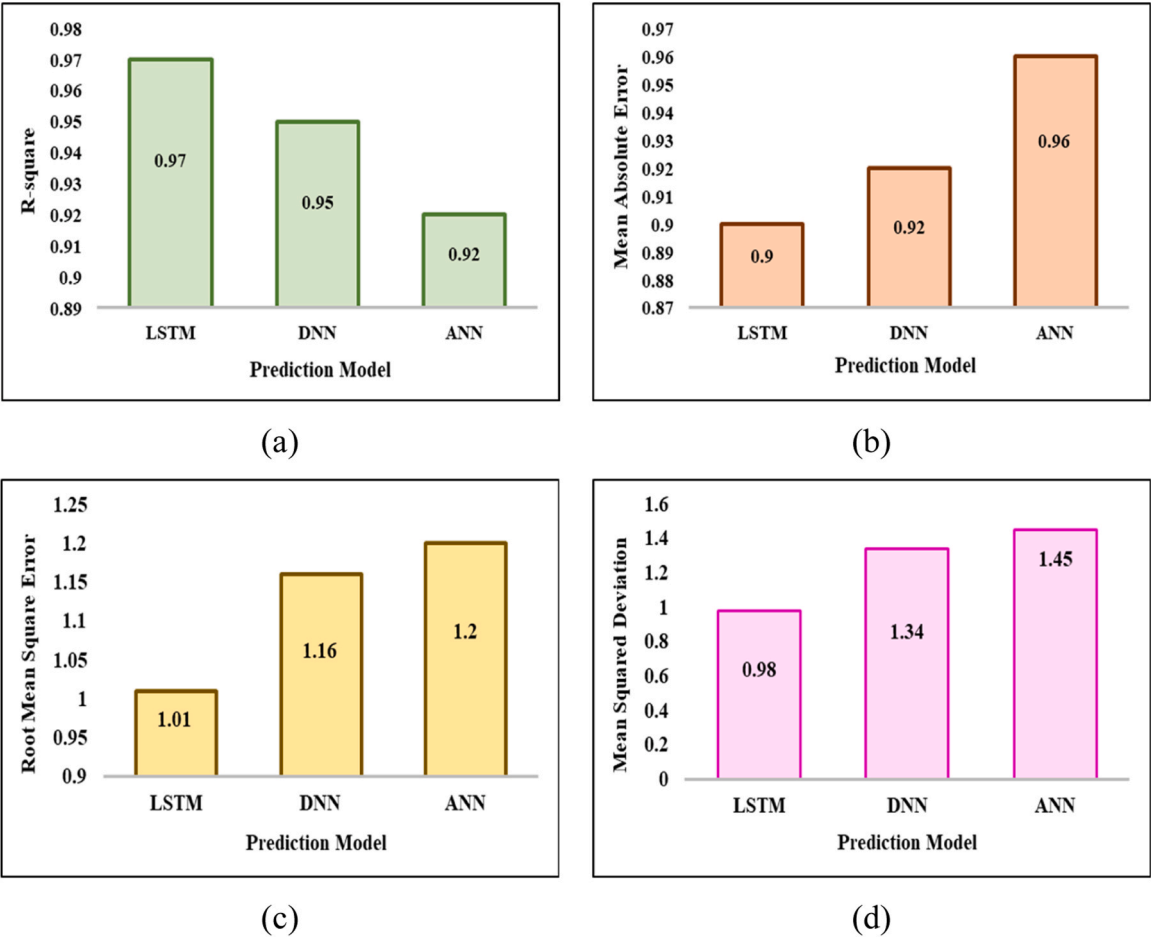


Fig. 7. Computational performance comparison (a) R-square; (b) Mean absolute error; (c) Root mean square error and (d) Mean squared deviation.

4.1. Performance metrics

The computational performance of the proposed model was evaluated against DNN and ANN models. First, the efficiency of the proposed method was examined in terms of R-square, training time, and MAE by comparing the ground truth and forecasted values. The performance metrics are presented in Table 2.

During the training phase, the model was optimized to forecast outcomes efficiently. The duration required for the model to learn from pre-processed data is defined as the training time. Notably, the LSTM network demonstrated superior efficiency in training duration compared to DNN and ANN, requiring only 2.69 s for 500 test cases, in contrast to 5.26 s and 3.88 s for DNN and ANN, respectively. The model's ability to predict the dependent variable based on independent variables was quantified using the R-squared (R^2) statistic, reflecting the fit quality of the data. An R^2 value of one indicates perfect prediction accuracy, although lower values are possible. The LSTM model achieved an R^2 of 0.97, outperforming the DNN and ANN models, which recorded values of 0.95 and 0.92, respectively Table 3.

MAE was utilized to compute the average magnitude of discrepancies between predicted and observed values, with optimal performance indicated by scores approaching zero. The LSTM model exhibited superior accuracy, achieving a lower error score, with MAEs of 0.90, 0.92, and 0.96 for LSTM, DNN, and ANN, respectively. RMSE, representing the standard deviation of prediction errors, assesses prediction accuracy by calculating the Euclidean distance between calculated and actual values. The LSTM model reported an RMSE of 1.01, demonstrating higher accuracy compared to the DNN and ANN models, which yielded RMSE values of 1.16 and 1.20, respectively. MSD was utilized to

evaluate the predictor's quality by estimating the squared differences between estimated and actual values. Models with MSD values close to zero are considered more accurate. The LSTM model achieved an MSD of 0.98, indicating a near-zero error level, whereas the DNN and ANN models exhibited higher MSD values of 1.34 and 1.45, respectively. The box plot revealed negligible differences between the DNN- and ANN-based models.

The outcomes demonstrated notable correlations among the evaluated models, with the LSTM model specifically designed to forecast annual electricity usage. The efficacy of the LSTM model was established through comparative analysis with other deep learning architectures, including DNN and ANN, based on parameters such as mean square deviation (MSD), mean absolute error (MAE), and root-mean-square error (RMSE). Values approaching zero and an R-squared value nearing one indicated superior performance. Among the models assessed, the LSTM model emerged as the most proficient in estimating annual energy consumption within this study. While the DNN model, renowned for its accuracy in analysing extensive datasets, also exhibited commendable performance, it did not surpass the LSTM model in effectiveness.

The performance comparison of the proposed model is articulated through a relative evaluation with LSTM, DNN, and ANN models, as depicted in Fig. 7. Fig. 7(a) presents the R^2 values computed by all three models, with LSTM demonstrating superior accuracy. Fig. 7(b) illustrates the MAE for each model, revealing that the proposed model achieves a lower error rate than its counterparts, confirming the precision of its energy prediction relative to actual energy consumption. Additionally, Figs. 7(c) and (d) detail the RMSE and MSD, respectively, further establishing the LSTM model's leading performance over the other models. Collectively, these results indicate that the energy usage

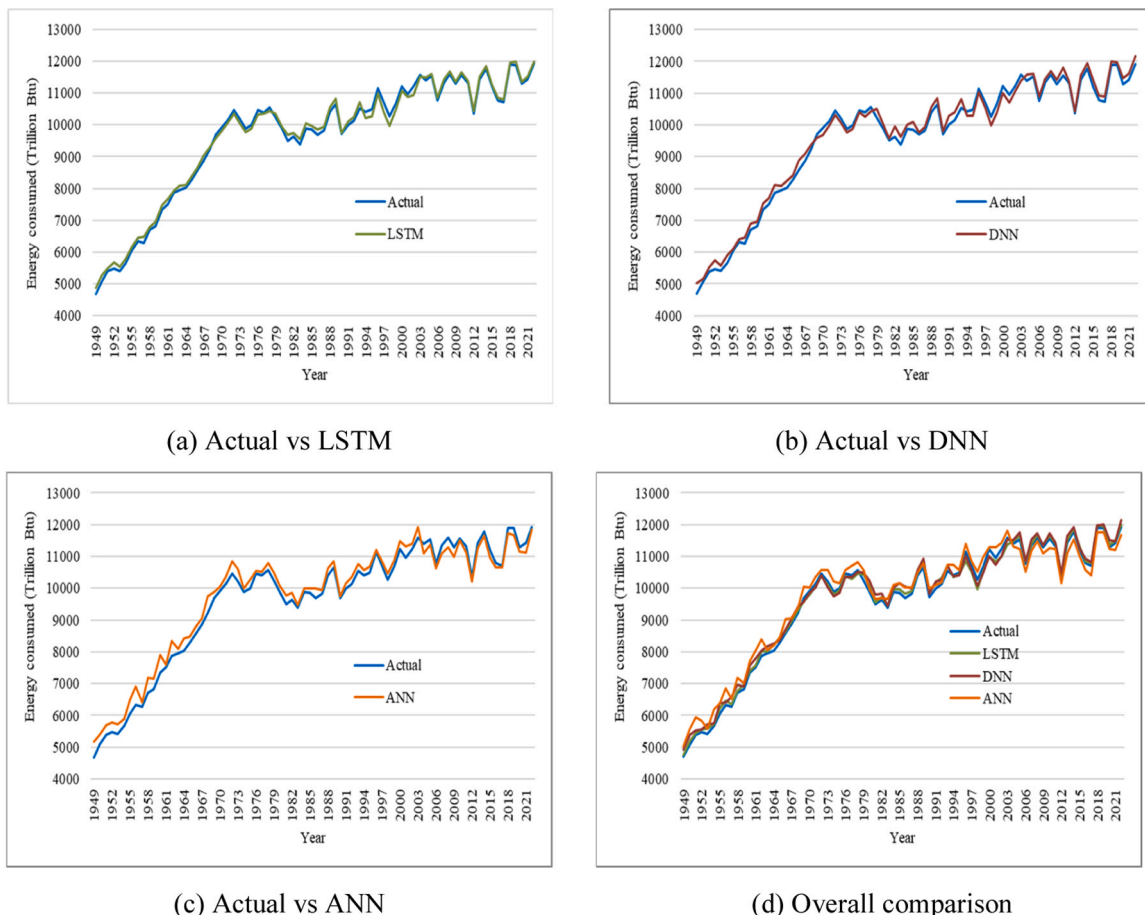


Fig. 8. Analysis of estimated values with actual values.

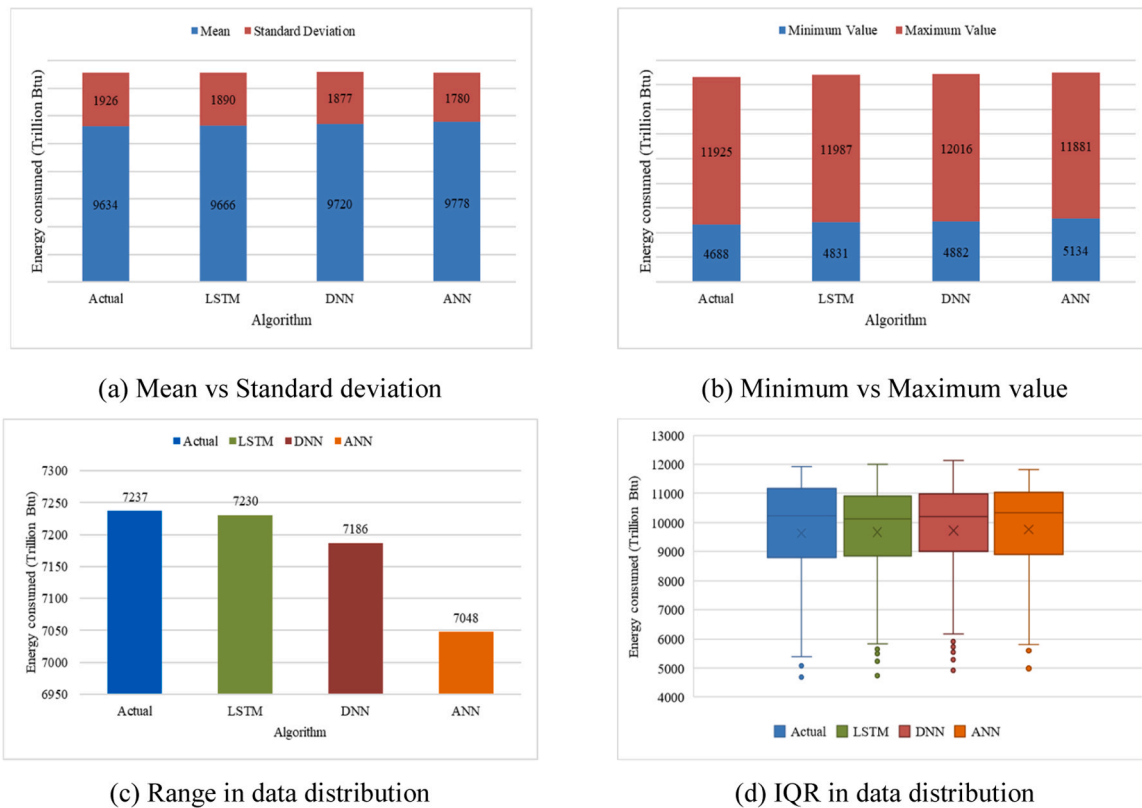


Fig. 9. Analysis of data distribution.

estimated by the proposed model closely approximates the actual energy consumed, outperforming the comparison methods in accuracy and reliability.

The performance evaluation of the proposed model undergoes rigorous validation against alternative models through a meticulous comparison of predicted energy usage with actual energy usage. For this assessment, annual energy consumption data from the dataset were utilized to gauge performance efficacy. Fig. 8(a) delineates the prediction performance of the LSTM model, showcasing its proficiency in forecasting energy consumption. Correspondingly, Figs. 8(b) and (c) illustrate the predictive accuracies of the DNN and ANN models, respectively, providing a comprehensive overview of each model's capability in energy prediction. The comparative analysis is further elaborated in Fig. 8(d), where the prediction accuracies of these models are juxtaposed with the actual values. This comparison distinctly reveals that the LSTM model's energy prediction not only surpasses that of the DNN and ANN models but also underscores the LSTM's superior methodological advantage in capturing the complexities of energy consumption patterns.

The effectiveness of the proposed method was rigorously assessed by analyzing the distribution of predicted values, as depicted in Fig. 9. Fig. 9(a) details the mean and standard deviation of the energy consumption for the LSTM model compared to other models. The proposed LSTM model's predictions, in terms of mean and standard deviation, closely align with the actual values, demonstrating superior accuracy over alternative approaches. Fig. 9(b) assesses the models' performance based on the predicted minimum and maximum energy values. Figs. 9(c) and (d) further illustrate the range of data distribution and the dispersion within the interquartile range for the models under comparison. These parameters reveal that the data distribution of the LSTM model closely mirrors the actual energy consumption, establishing its superiority over other evaluated methods.

The methodology includes a rigorous validation process where the performance of the LSTM model is compared against DNN and ANN

models. This comprehensive analysis confirms the effectiveness of the approach and highlights the specific conditions under which LSTM outperforms other models. Such detailed comparative studies are scarce in the current literature, making this contribution both unique and valuable for the advancement of building energy prediction research. The proposed predictive framework has significant practical implications for enhancing building energy efficiency. By providing accurate early-stage predictions of electricity consumption, the model enables more informed decision-making for energy management and policy development. This directly contributes to the broader goals of sustainability and energy conservation.

The proposed LSTM model offers significant practical applications in the field of energy consumption prediction. By accurately forecasting annual electricity usage, the model can inform energy management strategies and policy development. This early prediction capability is vital for integrating energy-efficient technologies and optimizing building designs before construction. Additionally, the LSTM model's ability to handle time-series data makes it adaptable for predicting energy consumption trends in the near future, facilitating proactive measures to address increasing energy demands and supporting sustainability goals.

5. Conclusions

Energy utilization and environmental conservation have become fundamental interests to society. Efficient utilization of available electrical energy is valued for both commercial and individual usage. With new technologies, perceptively planning the utilization of energy in the early stages of construction will significantly help the community. The proposed model can predict the electricity consumption of building structures before the construction stage to optimize the structure for efficient energy usage. A predictive model was proposed to predict the annual electricity consumption of residential buildings. The performance of the predictive model was compared with that of the DNN and

ANN models. The evaluation metrics, namely training time, R^2 , MAE, RMSE, and MSD, were used for performance evaluation and comparison. The results produced by LSTM showed a greater accuracy of 97 % and a lower error ratio than the DNN and ANN models. Furthermore, the predicted energy consumption values of the proposed model were evaluated against the actual values and the distribution of the data. LSTM has proven to be robust and reliable in predicting energy consumption.

While the LSTM model showed promising results, it is important to acknowledge potential limitations. The model's performance could be influenced by the quality and quantity of available data. For buildings with significantly different energy consumption patterns or lacking sufficient historical data, the model's accuracy may vary. Additionally, the LSTM model's reliance on specific climatic parameters means that extreme or atypical weather conditions could impact its predictions. Future research should focus on further validating the LSTM model's applicability across a wider range of building types and geographic locations. Incorporating additional variables, such as occupancy patterns and real-time energy usage data, could enhance the model's accuracy and generalizability. Real-world applications of the LSTM model in energy management systems can provide valuable insights and feedback for continuous improvement.

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Author Agreement Statement

We confirm that the revised manuscript “Advancing Building Energy Efficiency: A Deep Learning Approach to Early-Stage Prediction of Residential Electric Consumption” has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the sole contact for the Editorial process. He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

CRediT authorship contribution statement

Yuvaraj Natarajan: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization. **Sri Preethaa K R:** Writing – review & editing, Writing – original draft, Validation, Data curation, Conceptualization. **Karthic Sundaram:** Writing – review & editing, Writing – original draft, Visualization, Software, Formal analysis. **Akila Muthuramalingam:** Supervision, Project administration, Methodology, Formal analysis. **Ahmed Abdi Yusuf Ali:** Writing – review & editing, Writing – original draft, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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